AWS Production Infrastructure

Project Documentation

High Availability VPC with Auto Scaling & Application Load Balancer

Region: Asia Pacific (Mumbai) — ap-south-1

Date: August 1, 2026

AWS Account: 742851378637

1. About the Project

This project demonstrates how to create a Virtual Private Cloud (VPC) designed for a production environment on AWS. To improve resiliency, servers are deployed across two Availability Zones using an Auto Scaling Group and an Application Load Balancer.

For additional security, the servers are deployed in private subnets and only receive traffic through the load balancer. Outbound internet access for the private servers is provided via NAT Gateways, which are also deployed redundantly across both Availability Zones to maximize resiliency.

Key Design Goals

- High availability across two Availability Zones (ap-south-1a and ap-south-1b)
- Security through network isolation — application servers run in private subnets
- Elastic scalability using an Auto Scaling Group with CPU-based target tracking
- Centralized traffic distribution via an internet-facing Application Load Balancer
- Controlled internet access using NAT Gateways instead of public IPs on servers

2. Architecture Diagram

The diagram below illustrates the overall network and compute architecture, showing the VPC, public and private subnets, NAT Gateways, Application Load Balancer, Auto Scaling Group, and Security Groups across two Availability Zones.

About the Project:-

This example demonstrates how to create a VPC that you can use for servers in a production environment. To improve resiliency, you deploy the servers in two Availability Zones, by using an Auto Scaling group and an Application Load Balancer. For additional security, you deploy the servers in private subnets. The servers receive requests through the load balancer. The servers can connect to the internet by using a NAT gateway. To improve resiliency, you deploy the NAT gateway in both Availability Zones

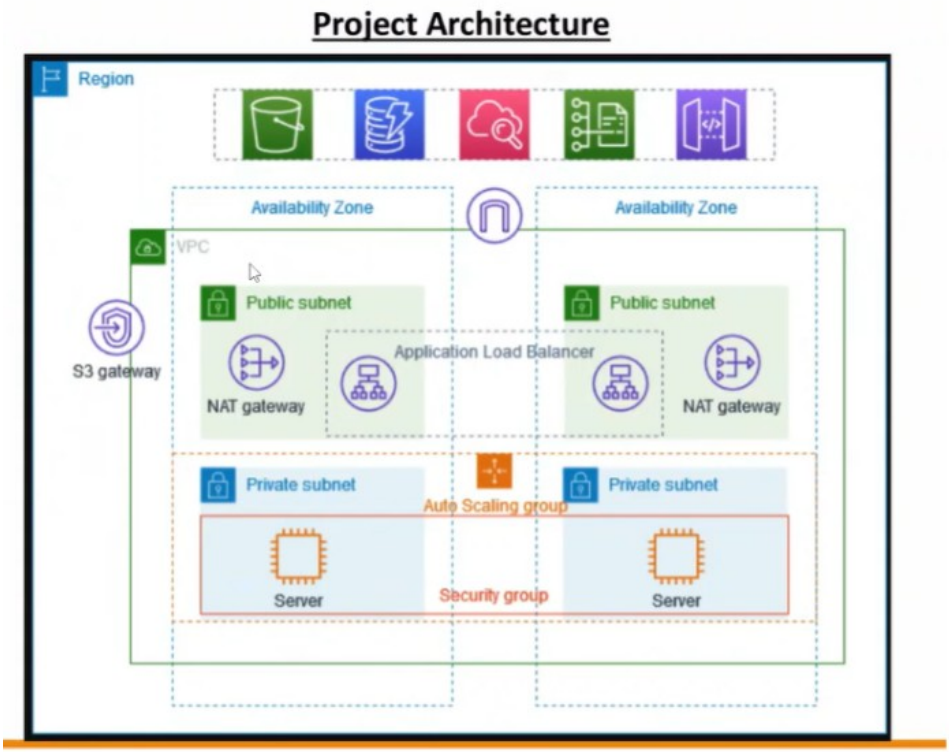


Figure 1: High-level project architecture

3. VPC & Networking Configuration

3.1 VPC Overview

Attribute	Value
VPC Name	production-vpc
VPC ID	vpc-02180accebcd083b3
Region	Asia Pacific (Mumbai) — ap-south-1

3.2 Subnets

The VPC contains four subnets — two public and two private — spread across two Availability Zones for high availability.

Subnet Name	CIDR Block	Subnet ID	Availability Zone
production-subnet-public1-ap-south-1a	10.0.0.0/20	subnet-01c1945d5d6b93e22	ap-south-1a
production-subnet-public2-ap-south-1b	10.0.16.0/20	subnet-02f824e7f3ce96dad	ap-south-1b
production-subnet-private1-ap-south-1a	10.0.128.0/20	subnet-0bdf61254c4079783	ap-south-1a
production-subnet-private2-ap-south-1b	10.0.144.0/20	subnet-0cf9b0c1369e64c93	ap-south-1b

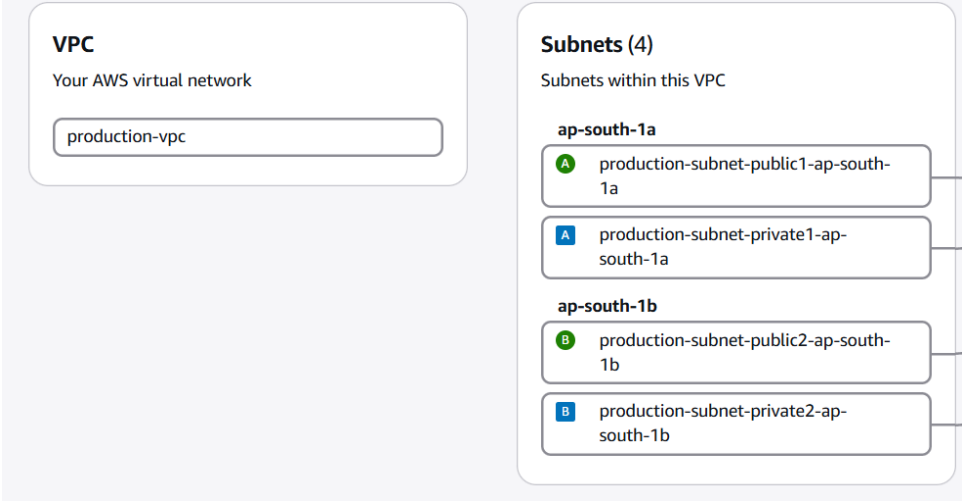


Figure 2: VPC resource map — VPC and Subnets

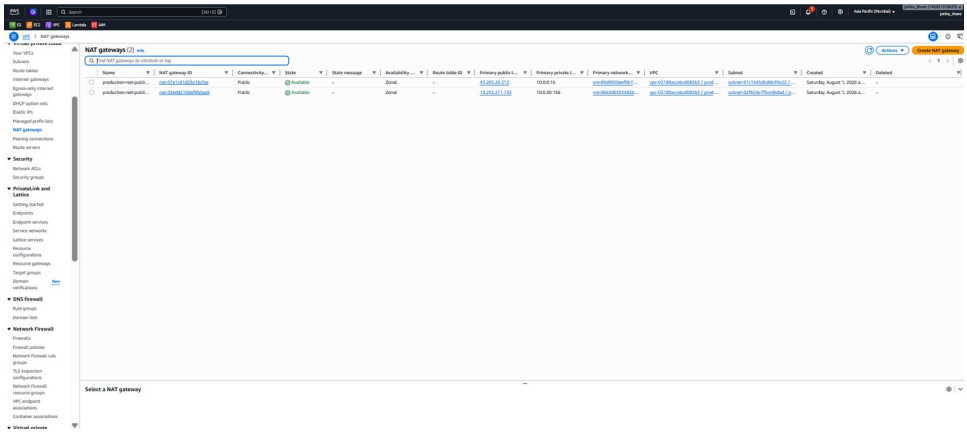


Figure 3: AWS Console — Subnets list

3.3 Route Tables & Network Connections

Four route tables manage traffic routing for the public and private subnets, working alongside the following network connections:

- production-igw — Internet Gateway for public subnet internet access
- production-nat-public1-ap-south-1a & production-nat-public2-ap-south-1b — NAT Gateways for private subnet outbound access
- production-vpce-s3 — VPC Endpoint providing private connectivity to Amazon S3

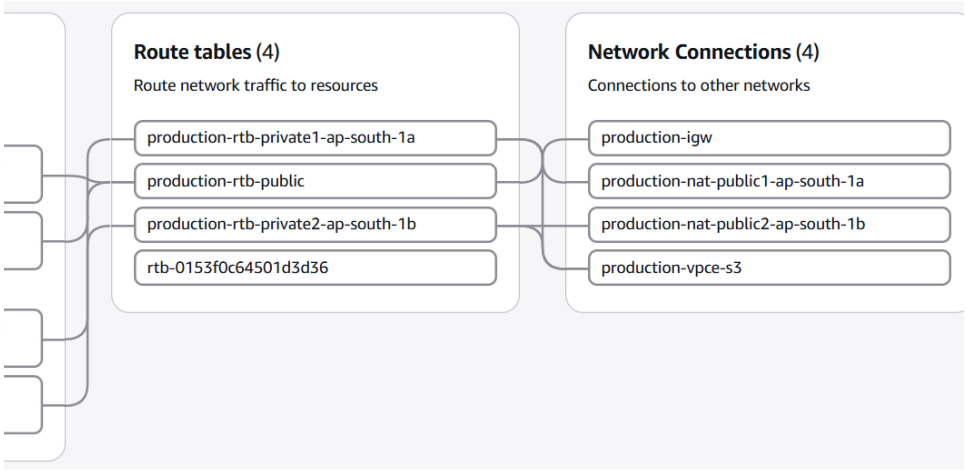


Figure 4: VPC resource map — Route Tables and Network Connections

3.4 NAT Gateways

Two NAT Gateways are deployed — one per Availability Zone — to ensure private subnet servers retain outbound internet connectivity even if one AZ experiences an outage.

Name	NAT Gateway ID	Public IP	Subnet
production-nat-public1	nat-07e1c61d23e16cfae	43.205.20.212	public1-ap-south-1a
production-nat-public2	nat-03edd210def8b0aa9	13.233.211.133	public2-ap-south-1b

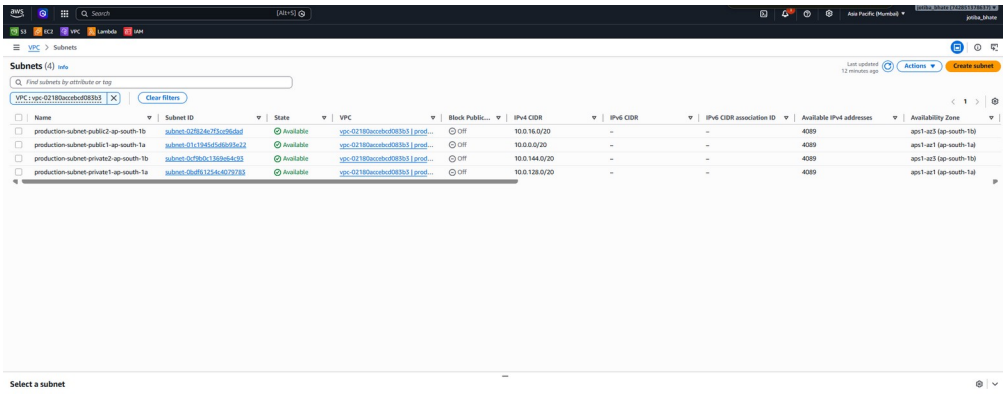


Figure 5: NAT Gateways — Available in both Availability Zones

4. Security Configuration

Two security groups control traffic flow within the VPC, following the principle of least privilege — the ALB accepts public HTTP traffic, while EC2 instances only accept traffic originating from the ALB.

Security Group Name	Security Group ID	Purpose
alb-sg	sg-07358a30d991ad8d0	Allows inbound HTTP (80) traffic from the internet to the ALB
private-ec2-sg	sg-076926f515e0e72f5	Allows inbound traffic to EC2 instances only from alb-sg

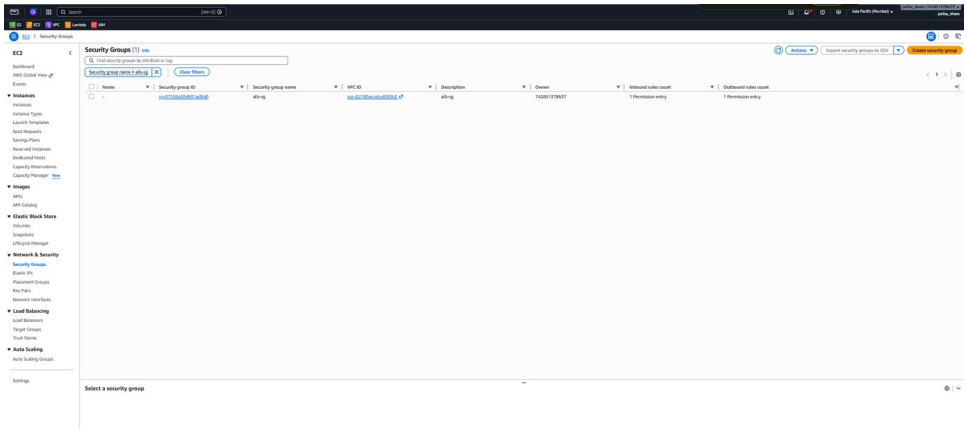


Figure 6: Security Group — alb-sg

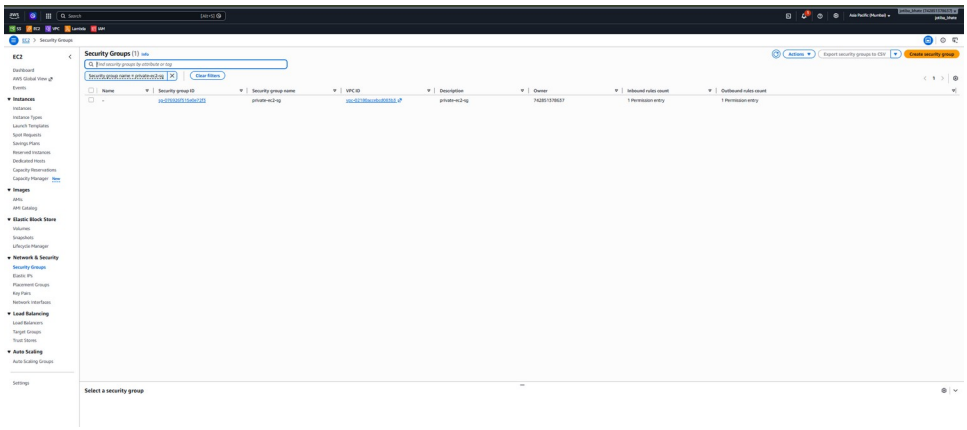


Figure 7: Security Group — private-ec2-sg

5. Compute Configuration

5.1 Launch Template

A launch template standardizes the configuration used by the Auto Scaling Group to launch new EC2 instances.

Attribute	Value
Launch Template Name	production-web-template
Launch Template ID	lt-06d356cb9d34f4967
Default Version	2
AMI ID	ami-01a00762f46d584a1
Instance Type	t3.micro
Key Pair Name	aws
Security Group	sg-076926f515e0e72f5 (private-ec2-sg)

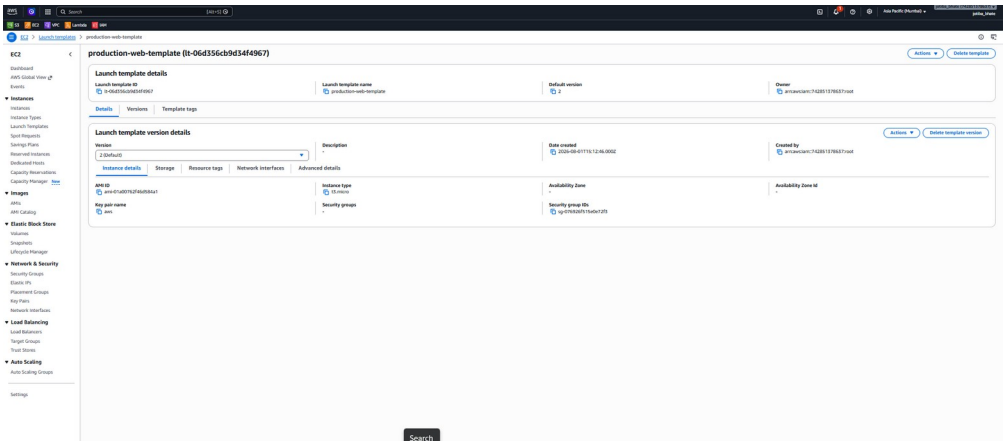


Figure 8: Launch Template — production-web-template

5.2 Auto Scaling Group

The Auto Scaling Group launches EC2 instances into the private subnets across both Availability Zones, maintaining a balanced distribution and scaling capacity based on CPU utilization.

Attribute	Value
ASG Name	production-asg
Desired Capacity	2
Scaling Limits (Min - Max)	2 - 4
Availability Zones	ap-south-1a, ap-south-1b
Subnets	subnet-0bdf61254c4079783, subnet-0cf9b0c1369e64c93
AZ Distribution	Balanced best effort
Status	At desired capacity

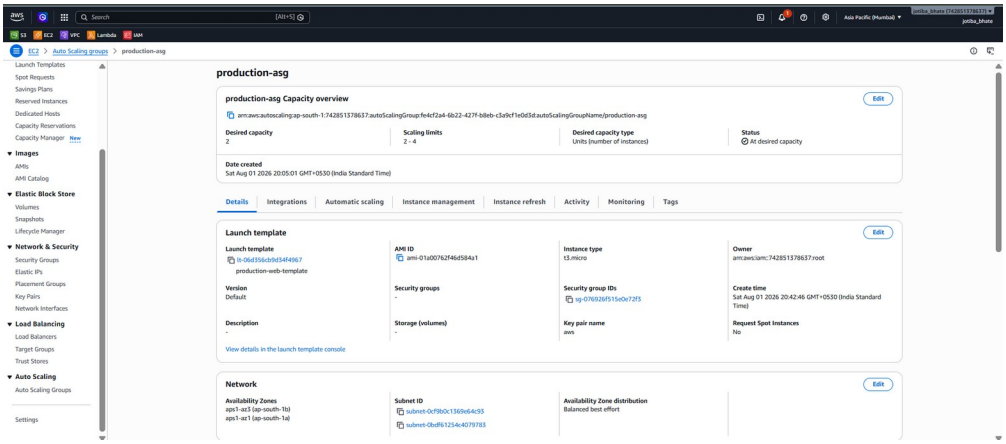


Figure 9: Auto Scaling Group — production-asg configuration

5.3 Instance Access via SSM Session Manager

Rather than exposing SSH access on the private EC2 instances, access is provided securely through AWS Systems Manager (SSM) Session Manager, using the production-ec2-ssm-role IAM role attached to each instance. This avoids the need for SSH keys, bastion hosts, or open inbound ports.

Attribute	Value
Instance ID	i-07f51020ae3195cfd
VPC ID	vpc-02180accebcd083b3
Security Group	sg-076926f515e0e72f5 (private-ec2-sg)
IAM Role	production-ec2-ssm-role
SSM Agent Status	Online — Connected
Agent Version	3.3.4121.0

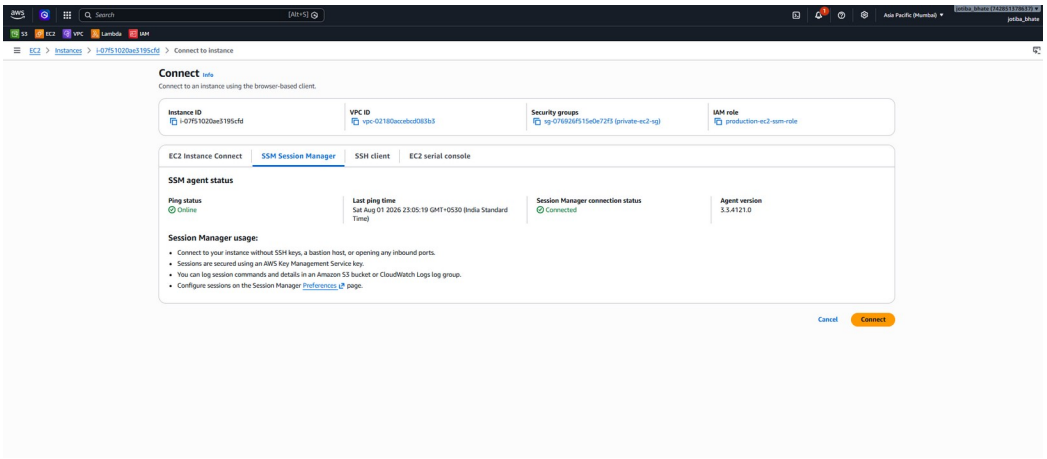


Figure 14: EC2 Connect — SSM Session Manager, agent online and connected

5.4 Load Testing (CPU Stress Simulation)

To validate that the Auto Scaling Group's target-tracking policies respond correctly to real load, a CPU stress test was run on a running instance via an SSM Session Manager shell, using the stress-ng utility (version 0.20.01).

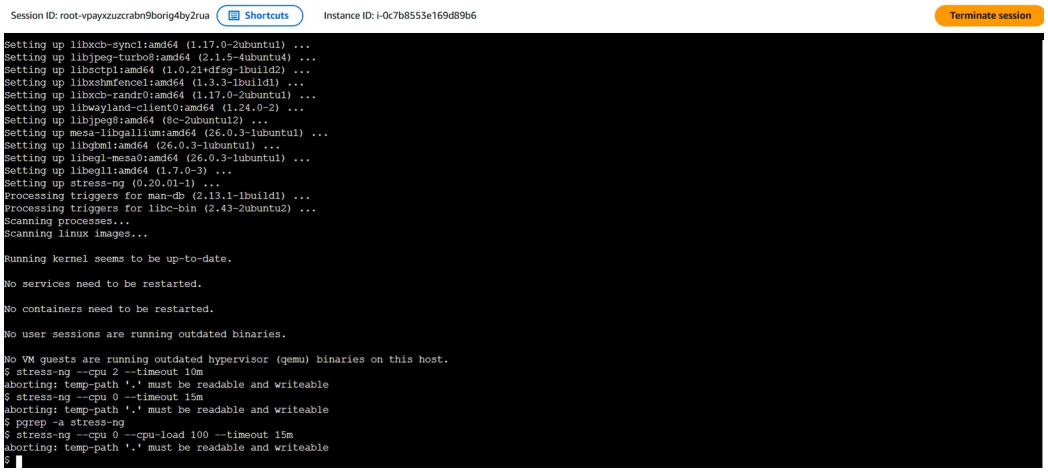


Figure 15: SSM Session Manager shell — installing and preparing stress-ng

The stress test was executed from the /tmp directory with a sustained 100% CPU load across 2 CPUs for a 15-minute duration, which drove average CPU utilization above the scale-up threshold and triggered the AlarmHigh target-tracking policy.

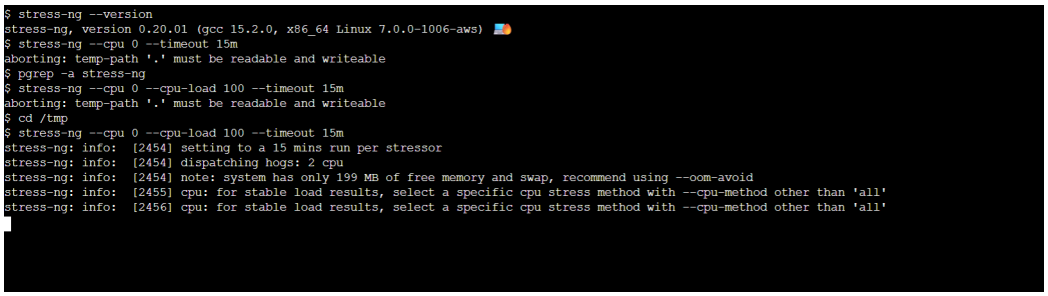


Figure 16: stress-ng running a sustained CPU load test (--cpu-load 100, --timeout 15m)

5.5 Auto Scaling Activity History

Target-tracking scaling policies (TargetTracking-production-asg-AlarmHigh and TargetTracking-production-asg-AlarmLow) automatically adjust capacity based on average CPU utilization. The activity history below confirms the policies functioned correctly — capacity scaled up from 2 to 4 instances under high CPU load, then back down to 2 as load decreased.

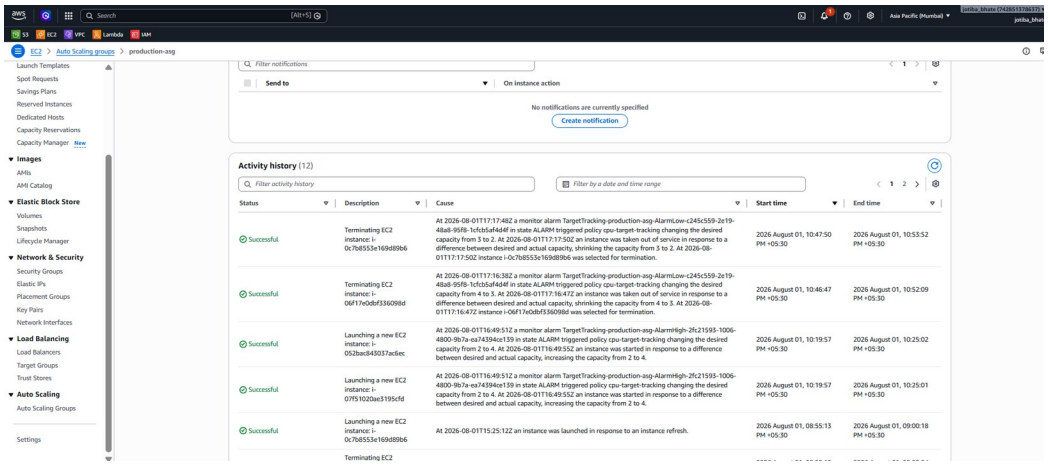


Figure 17: Auto Scaling Group — Activity history showing scale-up and scale-down events

6. Load Balancing

6.1 Application Load Balancer

Attribute	Value
Name	production-alb
Type	Application (Internet-facing)
VPC	vpc-02180accebcd083b3
Availability Zones	ap-south-1a, ap-south-1b
DNS Name	production-alb-1957332402.ap-south-1.elb.amazonaws.com
Status	Active
Listener	HTTP:80 → forwards to production-target-group (100%)

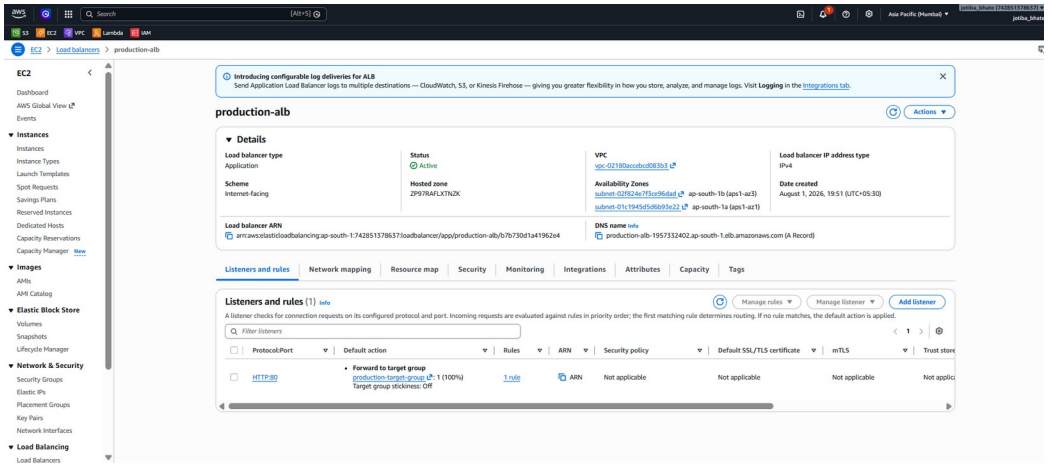


Figure 18: Application Load Balancer — production-alb

6.2 Target Group

Attribute	Value
Name	production-target-group
Target Type	Instance
Protocol : Port	HTTP : 80
Total Targets	2 (2 Healthy, 0 Unhealthy)

Instance ID	Port	Availability Zone	Health Status
i-07f51020ae3195cfd	80	ap-south-1b	Healthy
i-052bac843037ac6ec	80	ap-south-1a	Healthy

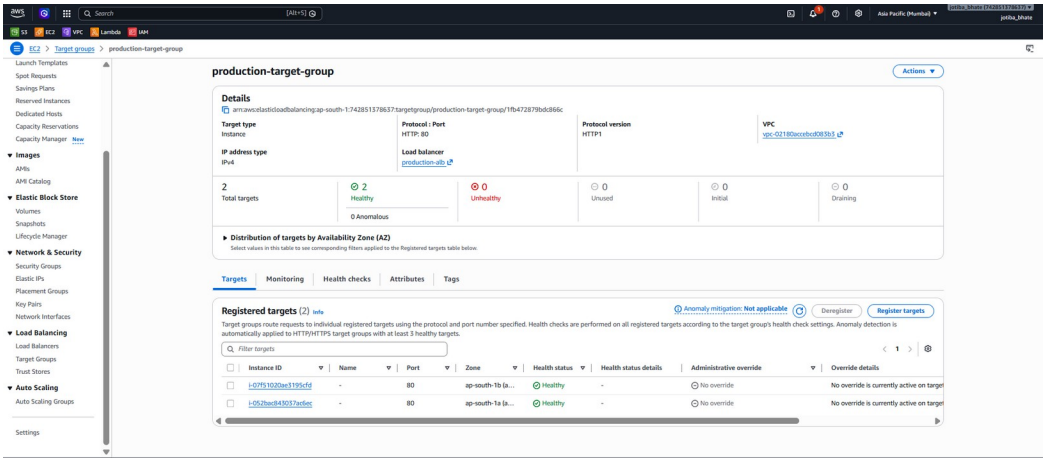


Figure 19: Target Group — production-target-group with healthy targets

7. Deployment Verification

The deployment was verified by accessing the Application Load Balancer's DNS name directly in a browser. The page confirms that the request was successfully routed through the ALB to a healthy EC2 instance running in a private subnet.

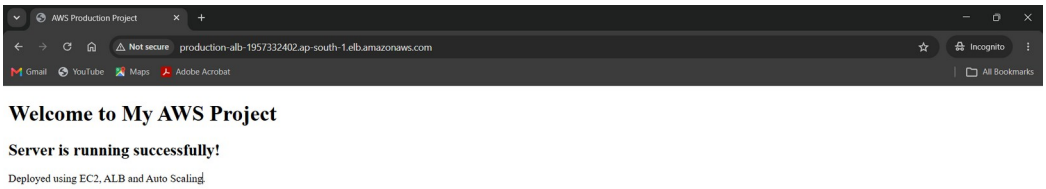


Figure 20: Browser test — application successfully served via the ALB

8. Summary

This project successfully demonstrates a production-grade, highly available AWS architecture using a custom VPC spanning two Availability Zones. Key outcomes include:

- A resilient network layer with public and private subnets, redundant NAT Gateways, and an S3 VPC endpoint
- Application servers isolated in private subnets, accessible only via the Application Load Balancer
- Automatic elastic scaling (2–4 instances) based on real-time CPU utilization
- Verified end-to-end connectivity from the public internet through the ALB to backend EC2 instances

This architecture provides a solid foundation for hosting production workloads with built-in fault tolerance, security isolation, and elastic scalability.